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Sourena Khanzadeh

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EDUCATION

Ph.D. in Computer Science

Sep 2023 – May 2026

Toronto Metropolitan University

Dissertation: Developing agentic formal verification methods for trustworthy smart contracts, with a focus on secure and correct gas optimization in Solidity. GPA: A+. Relevant coursework includes Heuristic Search, Cybersecurity, Deep Learning, and Directed Intelligent Robotic Systems.

Bachelor of Computer Science

Sep 2018 – Dec 2022

Toronto Metropolitan University

Relevant coursework includes Machine Learning, Artificial Intelligence, Reinforcement Learning, and Computer Vision, with grades of A+ achieved in each course.

WORK EXPERIENCE

Academic Assistant / Graduate Researcher

Jan 2023 – Apr 2026

Toronto Metropolitan University

Toronto

- Teaching Support and Instruction
 - Delivered weekly lab instruction and tutorials for undergraduate Computer Science courses, supporting cohorts of up to 180 students.
 - Mentored students in programming, algorithms, debugging, software design, and technical problem-solving.
 - Assisted with assignment design, grading, office hours, and student academic support.
 - Supported instruction in CPS125, CPS209, CPS506, CPS530, CPS706, and Career Co-op courses.
- Research
 - Conducted interdisciplinary research in Machine Learning, Deep Learning, Artificial Intelligence, Cybersecurity, and Blockchain systems.
 - Developed and evaluated deep learning models for intrusion-detection and adversarial cybersecurity settings, contributing to a peer-reviewed publication in International Journal of Information Security.
 - Contributed to technical writing, literature reviews, and research dissemination.

AI Research Scientist Intern

Jan 2024 – Jan 2025

National Research Council Canada

Toronto

- Conducted research and experimentation in deep learning and its applications to cybersecurity.
- Collaborated with interdisciplinary teams to prototype and evaluate novel approaches for intrusion detection systems and adversarial settings using reinforcement learning.
- Contributed to advancing the state of the art and supported research efforts leading to contributions to the computer science literature.

- Designed and developed fully functional full-stack systems in a collaborative, team-oriented environment.
- Worked closely with clients to understand requirements and deliver high-quality solutions.

SKILLS

Programming Python, C/C++, TypeScript, Java

AI/ML PyTorch, scikit-learn, Hugging Face, XGBoost, reinforcement learning, LLM agents

Systems/Infra Docker, Kubernetes, Redis, GitHub Actions, AWS

Databases MySQL, MongoDB, PostgreSQL

Web React, Angular, TailwindCSS

PUBLICATIONS & PROJECTS

Research on Blockchain

2023-2024

Gas Optimization

- S. Khanzadeh, N. Samreen and M. H. Alalfi, "Optimizing Gas Consumption in Ethereum Smart Contracts: Best Practices and Techniques," 2023 IEEE 23rd International Conference on Software Quality, Reliability, and Security Companion (QRS-C), Chiang Mai, Thailand, 2023, pp. 300-309, doi: 10.1109/QRS-C60940.2023.00056.

Research on Agents

2026

LLM Agents

- Khanzadeh, S., & Alalfi, M. H. (2026). GasZero: A Neuro-Symbolic Approach to Gas Optimization. Accepted to the 2026 IEEE International Computer Software and Applications Conference (COMPSAC).
- Khanzadeh, S., & Alalfi, M. H. (2026). Neuro-Symbolic Optimization: Agentic Formal Verification for Gas-Efficient Smart Contracts. Submitted to ACM Transactions on Software Engineering and Methodology (TOSEM) Special Issue on Agentic AI.

Research on Deep Learning

2024-2025

Deep Learning and Reinforcement Learning

- Khanzadeh, S., Neto, E.C.P., Iqbal, S. et al. An exploratory study on domain knowledge infusion in deep learning for automated threat defense. Int. J. Inf. Secur. 24, 71 (2025). <https://doi.org/10.1007/s10207-025-00987-4>
- Iqbal, S., Khanzadeh, S., Neto, E. C. P., Buffett, S., Sultana, M., & Taylor, A. (2025, October). Integrating Auxiliary Knowledge into Machine Learning to Improve the Detection of Cyberattacks. In 2025 International Symposium on Networks, Computers and Communications (ISNCC) (pp. 1-4). IEEE.
- Platnick, D., Khanzadeh, S., Sadeghian, A., & Valenzano, R. A. (2024). GANsemble for Small and Imbalanced Data Sets: A Baseline for Synthetic Microplastics Data. Canadian Artificial Intelligence Conference.

Research on Heuristic Search

2026

Search-Based AI

- Platnick, D., Tomasz, D., Earl, E., Khanzadeh, S., & Valenzano, R. (2026). Breadth-First Search vs. Restarting Random Walks for Escaping Uninformed Heuristic Regions. AAAI.